

Gate A review - Marketplace Orders

1. Unlock condition

Open only after the genuine Gate SQL attempt and evidence are saved. This review is diagnostic answer material, not a construction guide.

2. Core competency

Prove source grains and one-side keys; use INNER/LEFT deliberately; detect cardinality multiplication; protect the main-table measure before/after joining line rows; keep unmatched events visible; build readable staged SQL; validate windows/ranks/month comparisons; reconcile important totals.

3. Cardinality trap

Main rows = 30; line rows = 60; joined main-line rows = 60. Correct raw freight_amount total = 12,300.00. Naive post-line-join freight_amount total = 24,600.00. That inflated figure must be diagnosed, not accepted.

4. Critical failures

Unexplained result grain or multiplication; no one-side uniqueness check; DISTINCT used as a blind repair; hidden unmatched data; inflated coarse-grain measure; non-deterministic latest-row logic; broken LAG denominator; failed reconciliation; or unsupported business interpretation can block a pass.

Protected objective controls

Control	Expected
Main rows	30
Line rows	60
Event rows	9
Main x line joined rows	60
Main LEFT event rows	33
Raw line_sales total	624,888.00
Raw freight_amount total	12,300.00
Naive post-join freight_amount	24,600.00
Distinct primary dimension in main	8

5. Monthly line-measure controls

2026-01=127,411.20; 2026-02=70,391.70; 2026-03=107,847.00; 2026-04=120,862.80; 2026-05=79,239.60; 2026-06=119,135.70

6. How to diagnose a mismatch

Do not type the protected value into your query. Trace raw rows -> join counts -> one-side key proof -> line aggregation -> main-grain join -> monthly stage -> window/rank. Repair the earliest incorrect stage and rerun downstream controls.

Diagnostic route and retest state

SA1	INNER/LEFT choice, unmatched behavior, output grain.
SA2	Row multiplication, one-side uniqueness, inflated coarse-grain measure, reconciliation.
SA3-SA4	String standardization or date/month/elapsed-day logic.
SA5-SA6	Subquery result shape or CTE stage-grain/readability.
SA7-SA8	Window detail grain, partition validation, deterministic latest/rank or LAG denominator.
SA9	UNION vs UNION ALL semantics / awareness boundary.
M-SQLA	If multiple critical areas fail together, rebuild the integrated project evidence before a full retest.

7. Exact next route

Clean Gate A pass ends M04. Failed A -> Review A + targeted repair + changed evidence; Gate B only if explicitly assigned. Keep the original attempt, close this review, complete changed evidence, and append the new attempt.